

DELHI PUBLIC SCHOOL, BOKARO STEEL CITY

Half-Yearly Examination 2026 — Class X — Science (Physics)

Time Allowed: 90 minutes

Maximum Marks: 50

All questions are compulsory. Marks for each question are indicated against it. Write answers in the space provided on the answer sheet.

- Q1.** Define Newton's Second Law of Motion. [3 marks]
- Q2.** State the difference between speed and velocity. Give one example of each. [4 marks]
- Q3.** Explain the process of photosynthesis in green plants, and write the balanced chemical equation for it. [6 marks]
- Q4.** What is the SI unit of electric current? Define one ampere. [3 marks]
- Q5.** Name two renewable and two non-renewable sources of energy. [4 marks]

Section B — Numericals and Derivations

- Q6.** A car starts from rest and accelerates uniformly at 2 m/s^2 for 5 seconds. Calculate its final velocity and the distance covered. [5 marks]
- Q7.** Explain the difference between mitosis and meiosis. [5 marks]
- Q8.** State Ohm's Law. A resistor of 10Ω carries a current of 2 A. Calculate the potential difference across it. [4 marks]
- Q9.** What are the three states of matter? Describe how particle arrangement differs between them. [5 marks]
- Q10.** Define acceleration due to gravity. What is its approximate value near the Earth's surface? [3 marks]
- Q11.** Answer the following in the context of linear motion: [7 marks]
- (a) Derive the equation of motion $s = ut + \frac{1}{2}at^2$. [4 marks]
- (b) A car accelerates uniformly from rest at 2 m/s^2 . Find its velocity after 5 seconds. [3 marks]
- Q12.** Explain any two methods used to conserve water at the community level. [5 marks]